

Automated Supervisory Dispatch Control of a Solar–Grid–Electrolyzer Hydrogen Plant

A receding-horizon (MPC) controller for least-cost green-hydrogen production — validated against an ETAP digital twin and driven by real Ghardaïa weather and the Algerian time-of-use tariff.

PRESENTED BY

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Jury: **Prof. AKROUM Hamza (President)**
Mrs. MILOUDI Lalia (Examiner)

July 4, 2026

What we will cover

I

Context, problem & objectives

≈ 4 min

What green hydrogen is, why dispatch is a control problem, what we deliver.

II

System & methodology

≈ 8 min

A real-data plant model validated against the industry reference; then the optimiser and the controller.

III

Results

≈ 4 min

Daily dispatch, value vs traditional operation, MPC performance, full-year & storage.

IV

Deliverables, demo & conclusion

≈ 3 min

The live dashboard, contributions and future work.

WHO PRESENTS

BOUBIDI Youcef

The plant & the problem — context, real data, validated model, control formulation

BOUDJADJA Mohamed Akrem

The control & the payoff — PSO, MPC, results, live demo, conclusion

≈ 20 minutes · then questions



Context, Problem & Objectives

Casting least-cost hydrogen production as an automatic control problem.

Presented by BOUBIDI Youcef

Hydrogen: the densest clean fuel we can make

- An energy carrier, not a source: you make it, store it, ship it — and using it releases only water.
- The world already runs on it quietly: ≈ 97 Mt/yr for fertiliser (ammonia), refining and steel — $\sim 99\%$ of it made from fossil gas today (IEA).
- An electrolyzer makes the clean kind from water + electricity: it consumes $\approx 50\text{--}55$ kWh per kg (vs the 33.3 kWh a kg stores $\rightarrow \approx 60\%$ efficient).
- So electricity is roughly two-thirds of the cost of every green kilogram — the whole game is cheap electrons.

THE “COLOURS” OF HYDROGEN

Grey

from fossil gas · $\approx 9\text{--}10$ kg CO₂/kg (IEA)

Blue

grey + carbon capture

Green

renewables · ≈ 0 CO₂ — this thesis

ENERGY IN 1 KG OF FUEL — kWh (LOWER HEATING VALUE)

Hydrogen	33.3
Natural gas	13.9
Gasoline	12.1
Diesel	11.9

$\approx 3\times$ diesel, kilogram for kilogram — and the exhaust is water.

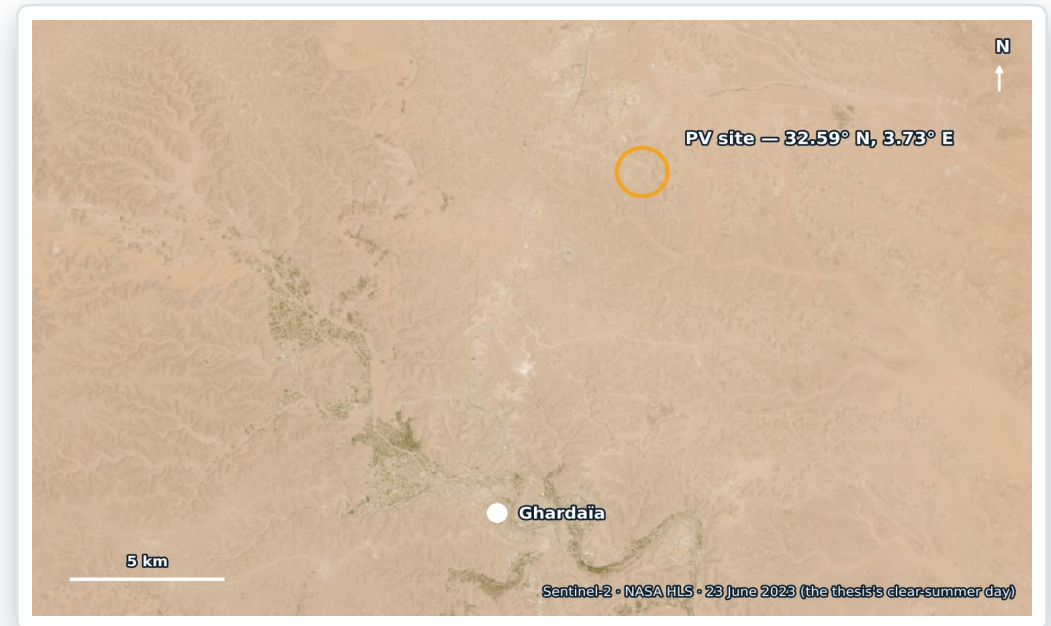
LHV data — U.S. DOE Hydrogen Program / standard engineering references.

Green hydrogen is **an electricity-cost problem $\rightarrow$ a timing problem $\rightarrow$ a control problem.**

The window: fossils pricing out, solar pricing in

- Fossil fuels are finite and increasingly carbon-priced: proven oil reserves ≈ 50 years at today's production rate (Energy Institute Statistical Review).
- From 2026 the EU carbon border tax (CBAM) makes exporters pay for embedded CO₂ — fossil-based products lose ground in Algeria's main export market.
- Meanwhile utility solar became the cheapest electricity in history: cost down $\approx 90\%$ since 2010 (IRENA) — but only when the sun shines.
- And clean-hydrogen demand is set to explode: ≈ 430 Mt by 2050 in the IEA net-zero pathway — over 4× today's market, nearly all of it low-carbon.

ALGERIA'S HAND — THE SITE FROM SPACE



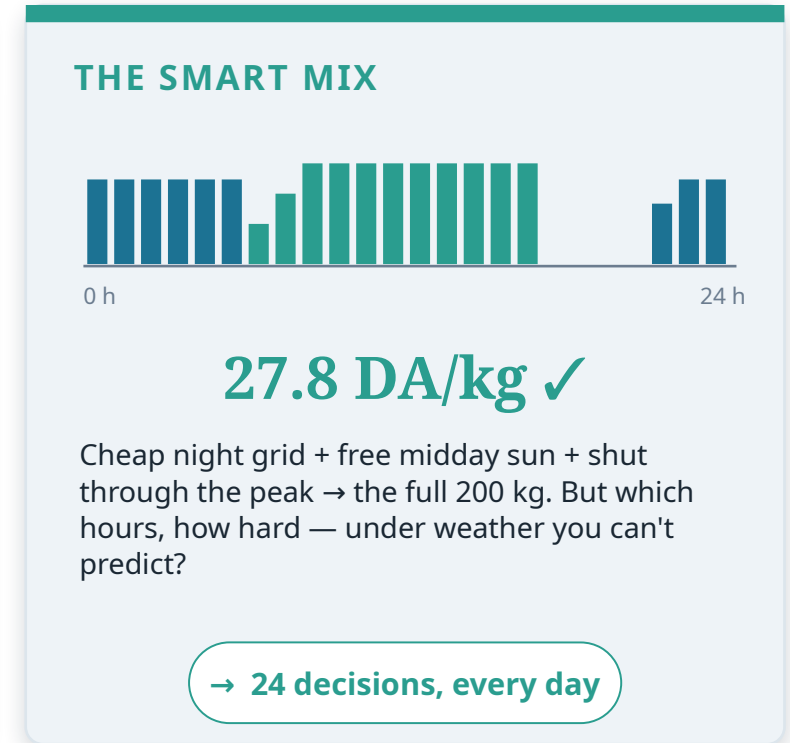
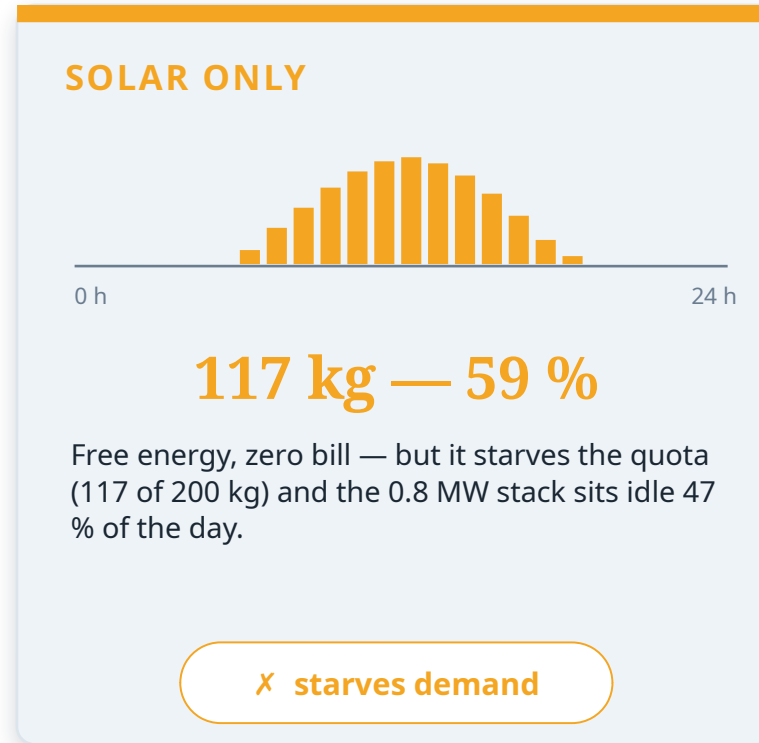
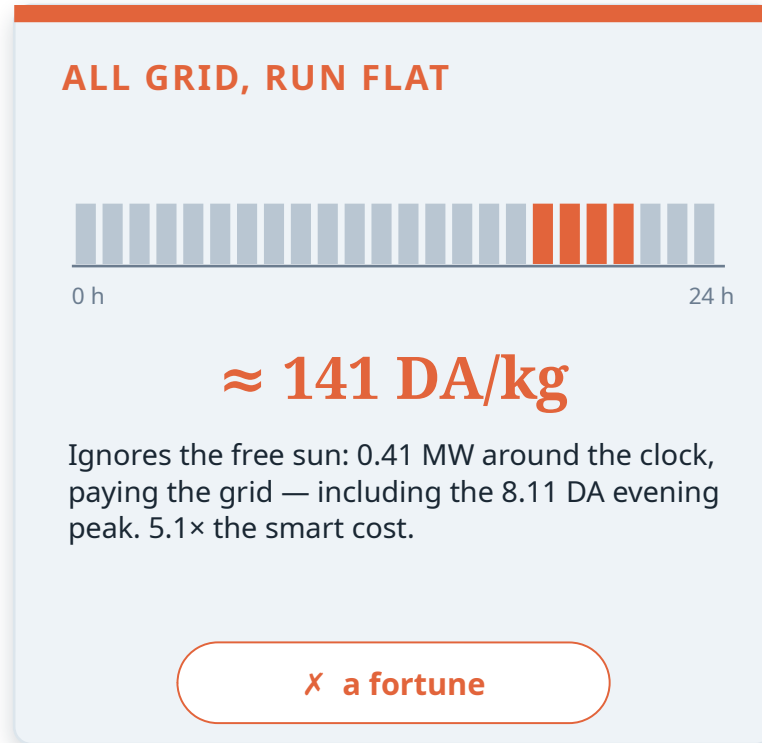
Sentinel-2 (NASA HLS), 23 June 2023 — the very day used as “clear summer” throughout this thesis.
Public domain.

☀️ **2,150 kWh/kWp/yr** · hydrocarbons $\approx 90\%$ of exports · **H₂ roadmap 2023**
· **SouthH2 → ambition $\approx 10\%$ of EU imports by 2040**

Fossils: finite, carbon-taxed · **Solar: -90% in a decade** → green H₂ turns desert sun into an exportable fuel.

Three ways to run the plant — two of them fail

Quota: 200 kg of H₂ per day. Real clear-summer 2023 day — every number computed with the validated plant model.



The real question: choosing those 24 setpoints — every day, automatically, under weather uncertainty, inside network limits. That is the control problem of this thesis.

Our Four Promises

1	A plant model you can trust The plant's electrical study, rebuilt in open, scriptable code — and proven to reproduce the industry-standard reference exactly.	24/24 checks pass
2	A controller that decides Dispatch cast as closed-loop supervisory control: measure, forecast, re-optimize — every hour, automatically.	24 decisions / day
3	Proof that it pays About half the cost of running the plant naively — and $\approx 98\%$ of what a perfect weather oracle could achieve.	–51 % vs reactive rule
4	Something you can touch An interactive dashboard on the validated engine — running live in this room.	1 live demo

The gap we fill: existing tools are accurate but manual, and dispatch studies rarely close the loop — ours is automated · network-checked · closed-loop, on real data.

II



System & Methodology

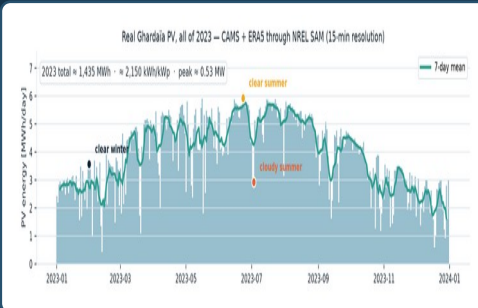
A validated digital twin on real data, an optimiser, and a feedback controller.

Presented by BOUBIDI Youcef

THE METHOD — ONE PICTURE

From real desert sun to least-cost hydrogen

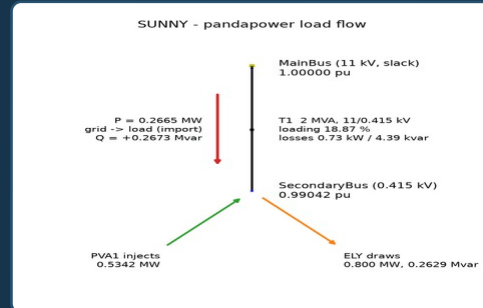
① REAL INPUTS



A full year of measured Ghardaia weather, and the official tariff.

real figure from this thesis

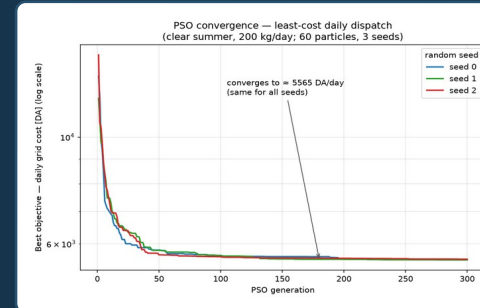
② A PLANT WE PROVED



The electrical model, validated 24/24 against the industry reference.

real figure from this thesis

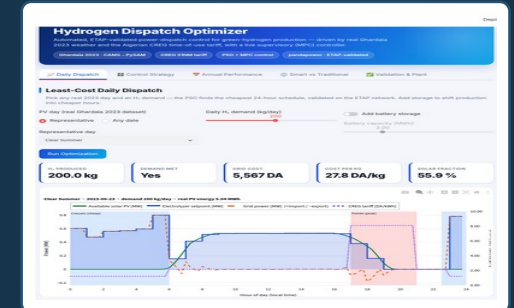
③ A CONTROLLER



Closed-loop dispatch: forecast, re-optimize, apply — every hour.

real figure from this thesis

④ PROOF, LIVE



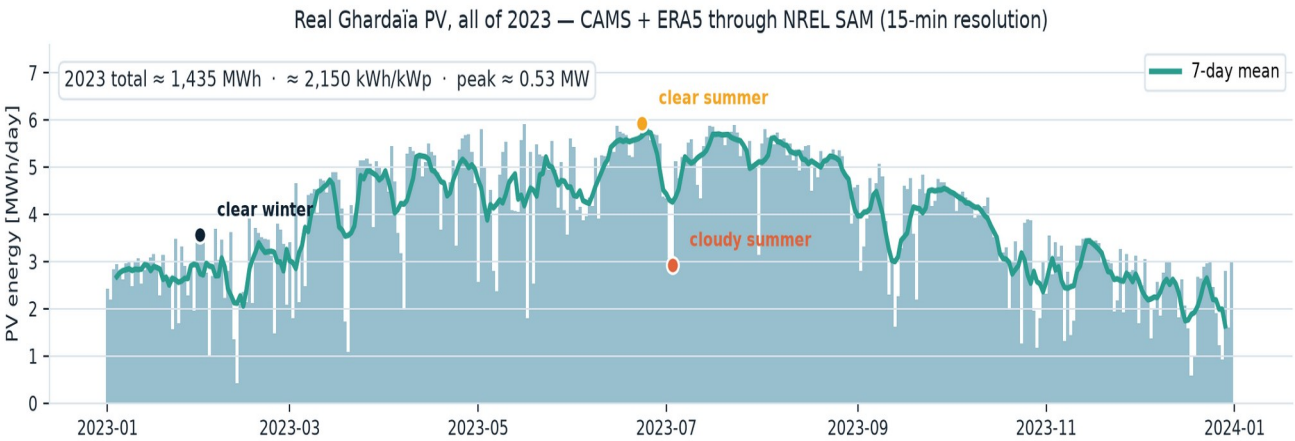
Every claim shown running — a live dashboard on the validated engine.

real figure from this thesis

Four stages, four slides ahead. Each tool is introduced the moment it does its work — and every figure you will see comes out of this pipeline.

Real Ghardaïa weather and the real Algerian tariff

- Solar: real CAMS satellite irradiance + ERA5 temperature/wind for Ghardaïa (32.59° N, 3.73° E), run through NREL's SAM (PySAM) → 15-minute AC power for all of 2023.
- Plant PV ≈ 669 kWp DC / 534 kW AC, single-axis tracking — matched to the ETAP interconnect; ≈ 2,150 kWh/kWp/yr, peak ≈ 0.53 MW.
- Prices: the official CREG / Sonelgaz medium-voltage time-of-use grid — three rates by hour of day (table →).



GRILLE TARIFAIRE MT — CREG / SONELGAZ · CODE 51NM

PÉRIODE	HEURES	DA/kWh
Pointe	17h00 – 21h00	8.1147
Pleines	06h00 – 17h00 21h00 – 22h30	2.1645
Creuses	22h30 – 06h00	1.2050
Prime de puissance (demand charge)		4.37 DA/kW/mois

Source : CREG, « Comment lire votre facture » — grille moyenne tension, code tarif 51NM.

Pointe = 6.7 × Creuses — the spread that makes timing pay.

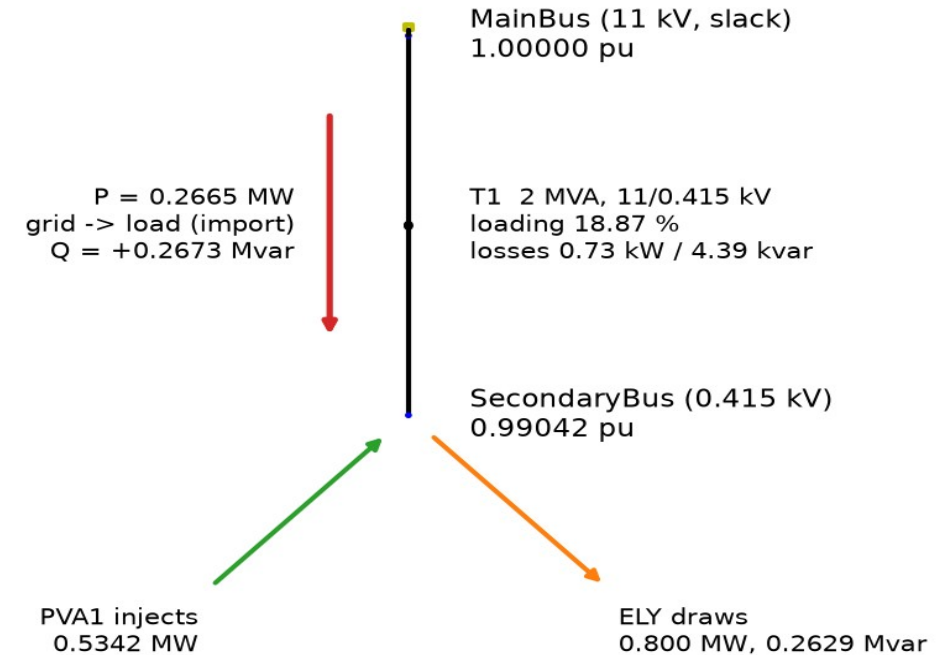
Prove the model before trusting it

- The plant's official electrical study lives in ETAP — the industry-standard power-system tool. Trusted, but GUI-bound: it cannot be scripted across a year of dispatch.
- So we rebuilt the same network in pandapower, an open-source load-flow engine we can automate: grid → 11 kV → 2 MVA transformer → 0.415 kV → PV + electrolyzer.
- Same Newton-Raphson load flow, three operating points (sunny / night / PV-export): 24/24 checks agree — voltages ± 0.01 pu, loading within 2 pts, flows within 2 %.
- The credibility anchor: everything downstream runs on a model that matches the professional reference — including the reverse-power-flow sign.

24 / 24

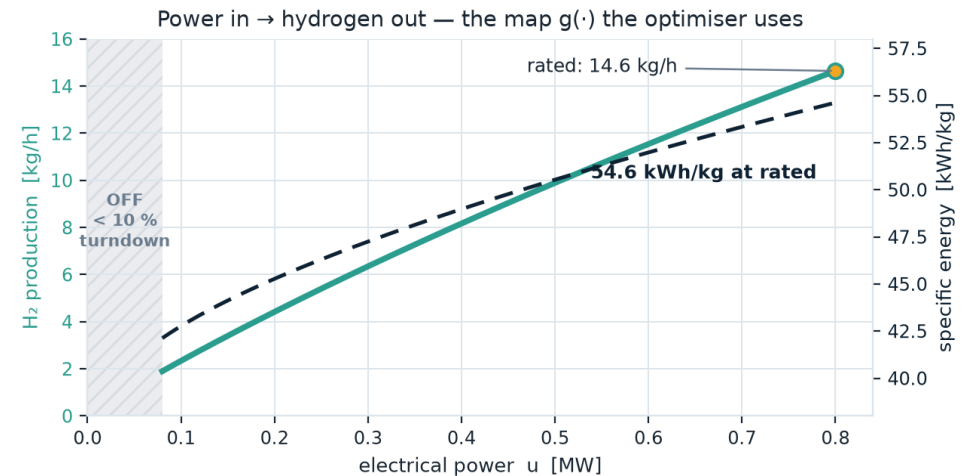
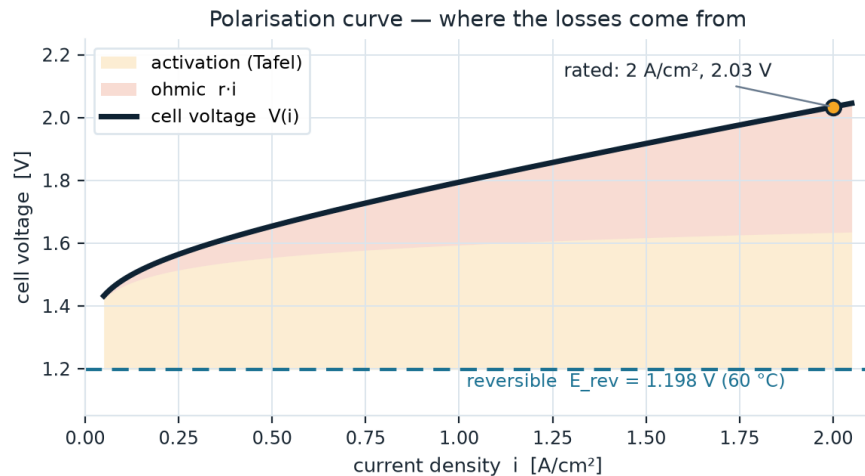
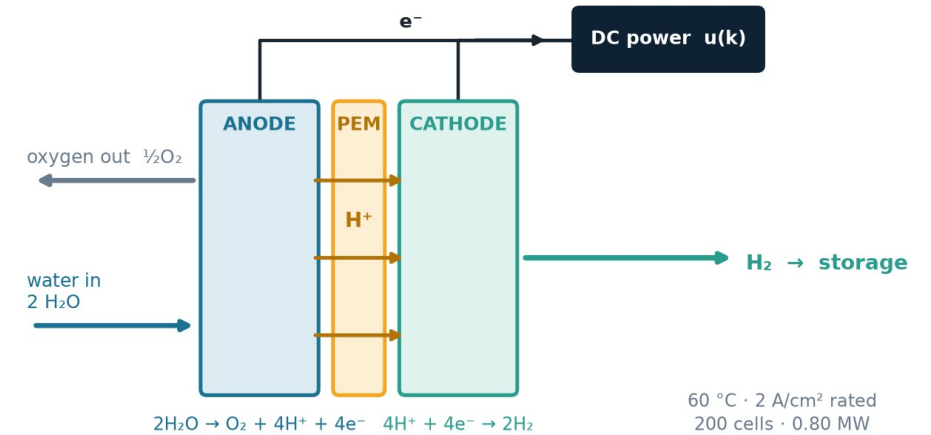
validation checks pass

SUNNY - pandapower load flow



PEM electrolyzer: power in, hydrogen out

- A PEM stack modelled from physics: reversible voltage + Tafel activation + ohmic loss give $V(i)$; Faraday's law turns current into hydrogen mass flow.
- Rated 0.800 MW at $\approx 2 \text{ A/cm}^2 \rightarrow \approx 14.6 \text{ kg H}_2/\text{h}$ at $\approx 54 \text{ kWh/kg}$; a 10 % minimum turndown (0.080 MW) — below it the stack switches off.
- The optimiser calls a fast lookup table of this exact curve, so a full year of dispatch is cheap to evaluate.



Dispatch written as a control problem

State time, battery charge, cumulative H₂ made

Control input electrolyzer setpoint (+ battery power)

Disturbance available PV — measured, forecast ahead

Constraints $V \in 0.95\text{--}1.05$ pu, transformer ≤ 100 %, demand, turndown

Cost functional daily grid energy cost (time-of-use)

WHY NO TEXTBOOK SOLVER FITS

Non-linear

the electrolyzer map $g(u)$ bends — efficiency shifts with load

Discontinuous

the tariff jumps between 3 rates — a 6.7× cliff at 17h00

Disjoint input

off, or 10–100 % — a hole in the feasible set

Uncertain

tomorrow's sun is unknown — plans go stale within hours

So the method has two parts: ② a solver that finds the cheapest possible day — then ③ a loop that survives imperfect forecasts.

The problem, formally

PLANT — DISCRETE-TIME STATE-SPACE

$$\mathbf{x}(k+1) = \mathbf{x}(k) + T \cdot \mathbf{g}(\mathbf{u}(k))$$

$$\mathbf{q}(k) = [\mathbf{u}(k) - \mathbf{d}(k)]^+ \quad \text{grid import — where } d \text{ acts}$$

$$\mathbf{y}(k) = \mathbf{x}(k) \quad \text{measured every hour}$$

x hydrogen produced so far today [kg] — one state

u electrolyzer setpoint $\in \{0\} \cup [0.08, 0.80]$ MW

d available PV [MW] — disturbance: enters q (the bill), not the dynamics

g(·) electrolyzer map: polarisation curve + Faraday

Hammerstein plant — locally an integrator: $\mathbf{G}(z) = \mathbf{K}/(z-1)$, $\mathbf{K} = T \cdot \mathbf{g}'(\mathbf{u}_0) \approx 15\text{--}22 \text{ kg/MW}$

OBJECTIVE — LEAST-COST DISPATCH

$$\min \quad J = \sum p(k) \cdot [\mathbf{u}(k) - \mathbf{d}(k)]^+ \cdot \Delta t$$

$= \sum p(k) \cdot q(k) \cdot \Delta t$ — pay only for net import $\cdot 96 \times 15\text{-min steps}$

$$\text{s.t.} \quad \mathbf{x}(24) \geq \mathbf{D}$$

terminal demand constraint — softened as $+ \lambda \cdot [D - x(24)]^+$, $\lambda = 10^7 \text{ DA/kg}$

$p(k) \in \{1.21, 2.16, 8.11\} \text{ DA/kWh}$ — the 3-rate CREG tariff (discontinuous in time)

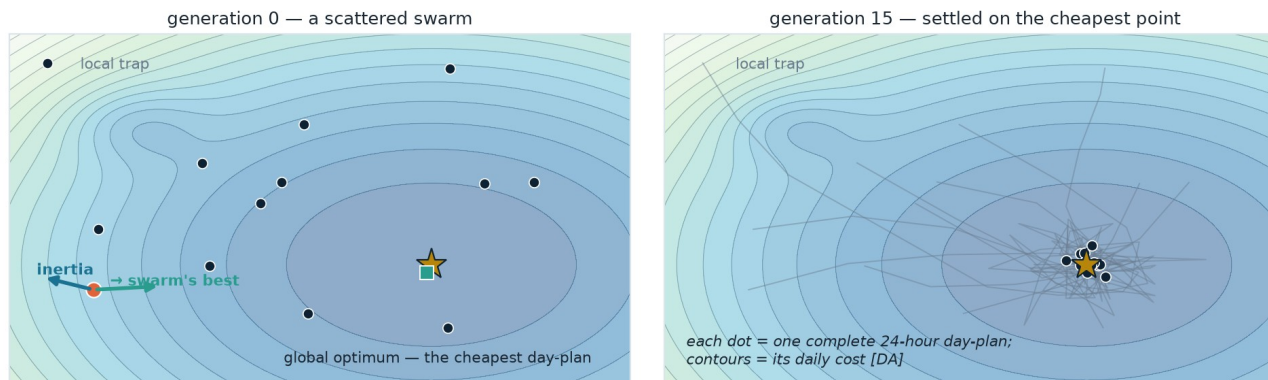
$V \in [0.95, 1.05] \text{ pu}$ · transformer $\leq 100\%$ — verified on the ETAP-validated load flow (never binding at this PV size)

Input set $\{0\} \cup [0.08, 0.80]$ is disjoint and g is non-linear $\rightarrow$ non-convex $\rightarrow$ solved with PSO

A bill, not a setpoint: the cost is money and the only reference is the end-of-day quota $x(24) \geq D$ — there is nothing to 'track'; the tariff itself shapes the optimal behaviour.

What is particle-swarm optimisation?

- A population-based optimiser inspired by how a flock searches for food: many candidate solutions explore the search space together.
- Each particle is one complete candidate day-plan, with a position (the solution itself) and a velocity (how it is changing).
- Every iteration a particle is pulled two ways — toward its own best find, and toward the whole swarm's best. That balance of exploring and exploiting converges on the global optimum.
- It is gradient-free: it needs no derivatives, so it copes with our non-linear electrolyzer and the discontinuous 3-rate tariff.



THE UPDATE, EACH ITERATION

$$\mathbf{v}_i \leftarrow w \cdot \mathbf{v}_i + c_1 r_1 (\mathbf{p}_i - \mathbf{x}_i) + c_2 r_2 (\mathbf{g} - \mathbf{x}_i)$$

$$\mathbf{x}_i \leftarrow \mathbf{x}_i + \mathbf{v}_i$$

$$w \cdot \mathbf{v}_i$$

inertia — keep exploring

$$c_1 r_1 (\mathbf{p}_i - \mathbf{x}_i)$$

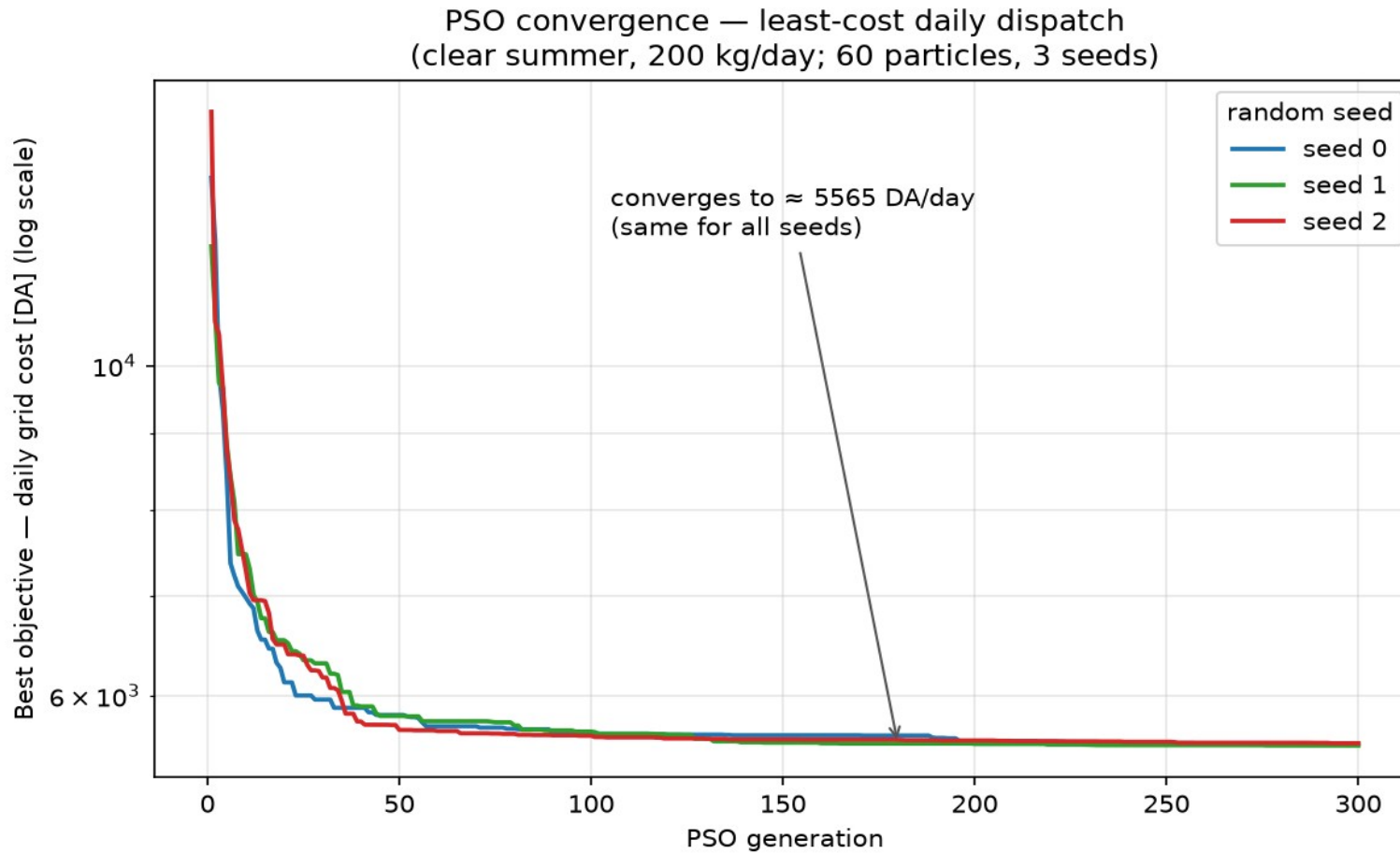
pull toward its personal best

$$c_2 r_2 (\mathbf{g} - \mathbf{x}_i)$$

pull toward the swarm's best

$r_1, r_2 \sim U(0,1)$, fresh each step · $|v|$ capped at 20 % of range · pymoo: $w = 0.9$, $c_1 = c_2 = 2.0$, adapted online.

PSO applied: how each daily plan is solved



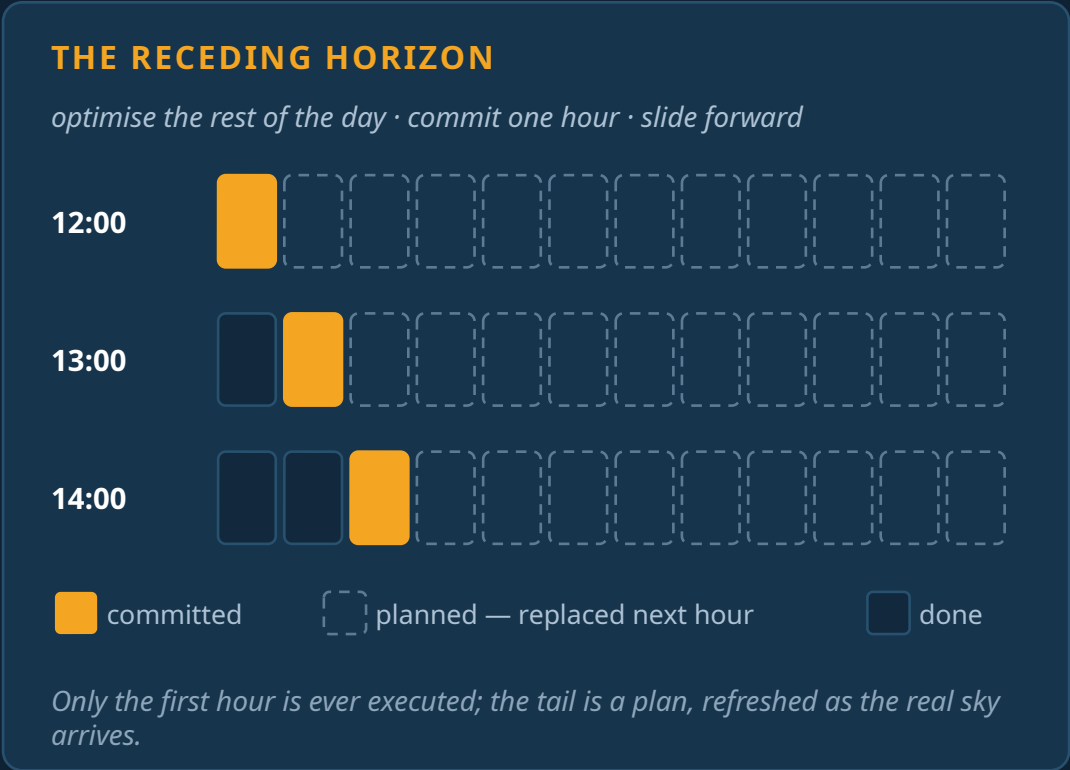
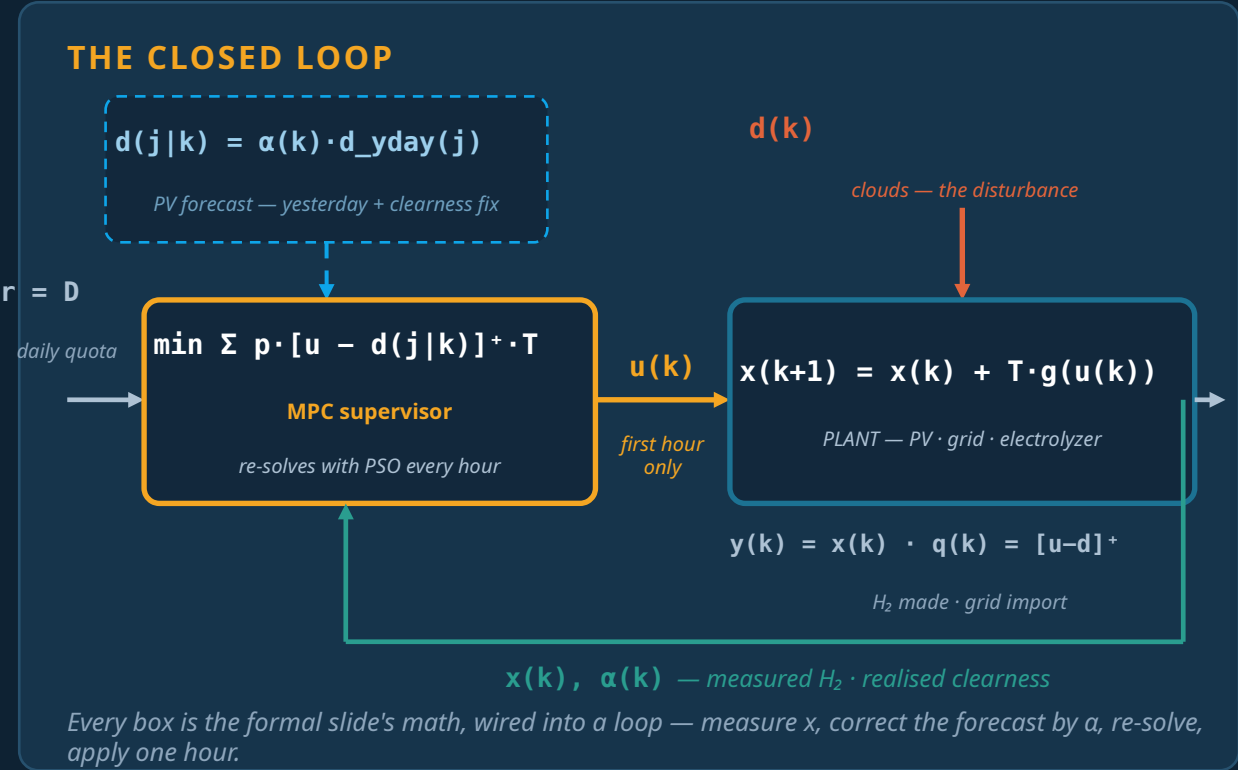
3 / 3

random seeds reach the same optimum

- The search is hard: 24 hourly setpoints, a non-linear electrolyzer, a discontinuous 3-rate tariff — no gradients to follow.
- PSO flies a swarm of ~ 60 candidate day-plans; each is pulled toward its own best and the swarm's best.
- Fitness = exactly the J of the formal statement: the 15-min grid bill + $10^7 \times$ any shortfall. Converges in ≈ 150 generations; 3 seeds, one optimum.
- But note what we assumed: the day's weather, known in advance. Tomorrow's never is — closing that gap is the loop's job, next.

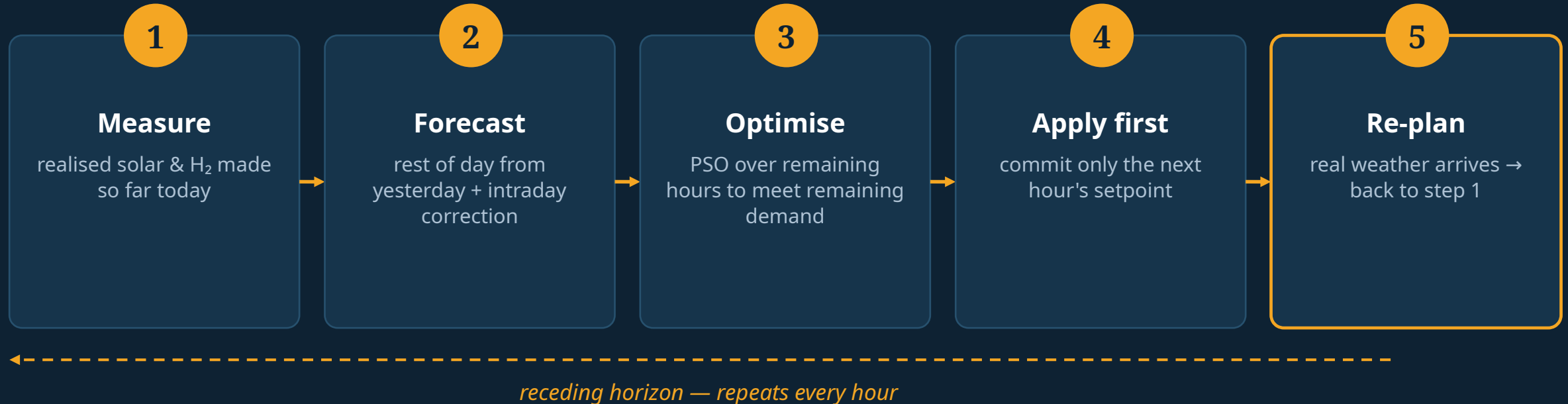
Model Predictive Control (MPC)

PSO runs inside this loop, ~24× a day — not “PSO then MPC”.



Supervisory, not low-level: the MPC sends one setpoint per hour; the electrolyzer's own fast controllers track it underneath — we command the plant, we don't micro-manage it.

Receding-horizon MPC: plan, act, re-plan



$$u^*(k) = \arg \min \sum p(j) \cdot [u(j) - d(j|k)]^+ \cdot T + \lambda \cdot [D - x(k) - \sum g(u(j)) \cdot T]^+$$

over $u(k) \dots u(23)$ — the horizon shrinks · $d(j|k) = \alpha(k) \cdot d_{\text{yesterday}}(j)$, α = today's realised clearness · apply only $u^*(k)$

$u(k) = \kappa(x(k), \alpha(k))$
an implicit feedback law

Benchmarks: perfect-foresight (unreachable lower bound) · reactive rule-based (time-blind constant rate)

III



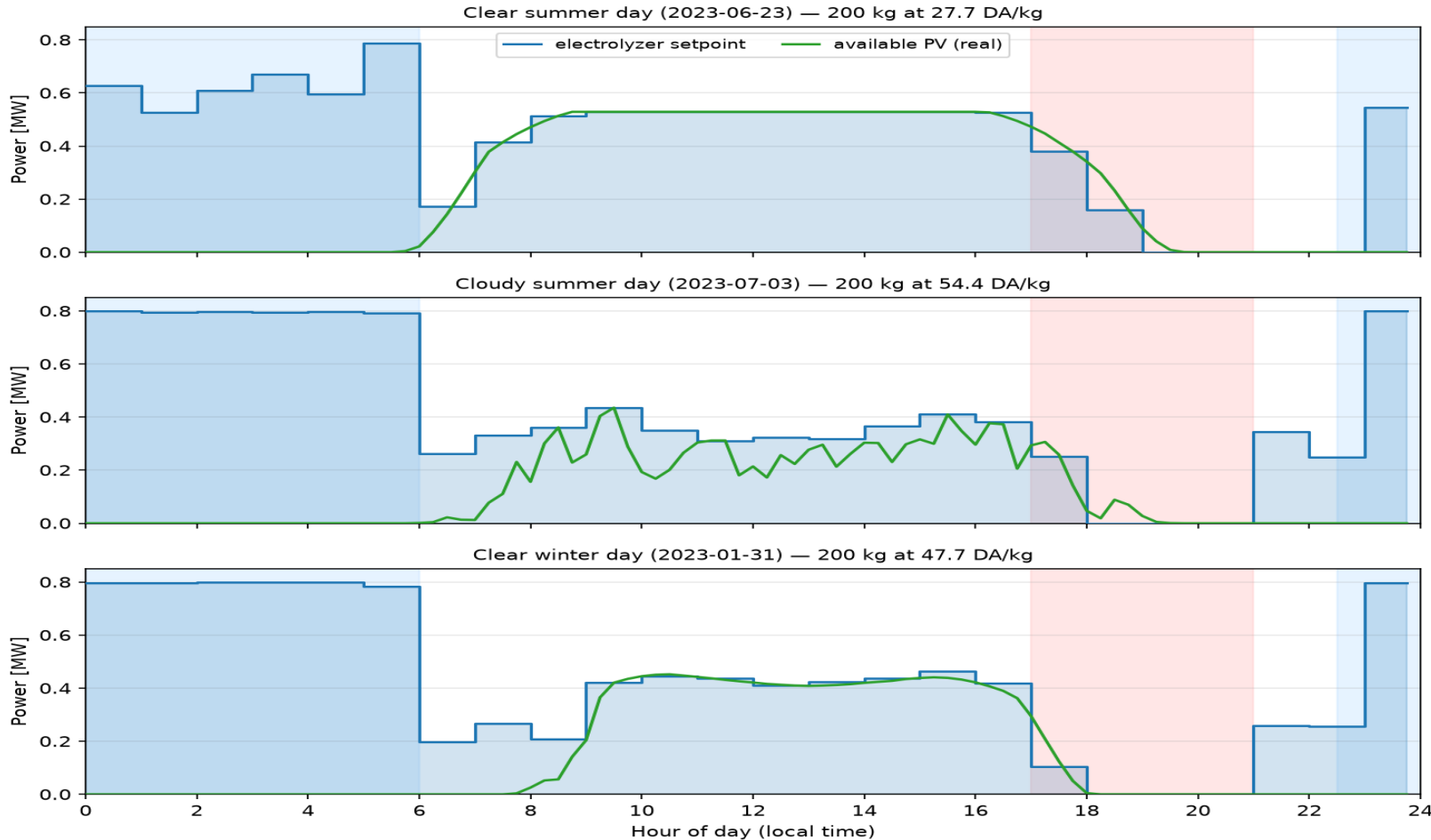
Results

What the controller does, and what it is worth — on real 2023 data.

Presented by BOUDJADJA Mohamed Akrem

It rides free solar and dodges the evening peak

Least-cost dispatch to meet 200 kg/day — Ghardaïa (real CAMS+PySAM, 15-min)
blue = off-peak nights, red = 17–21 h peak



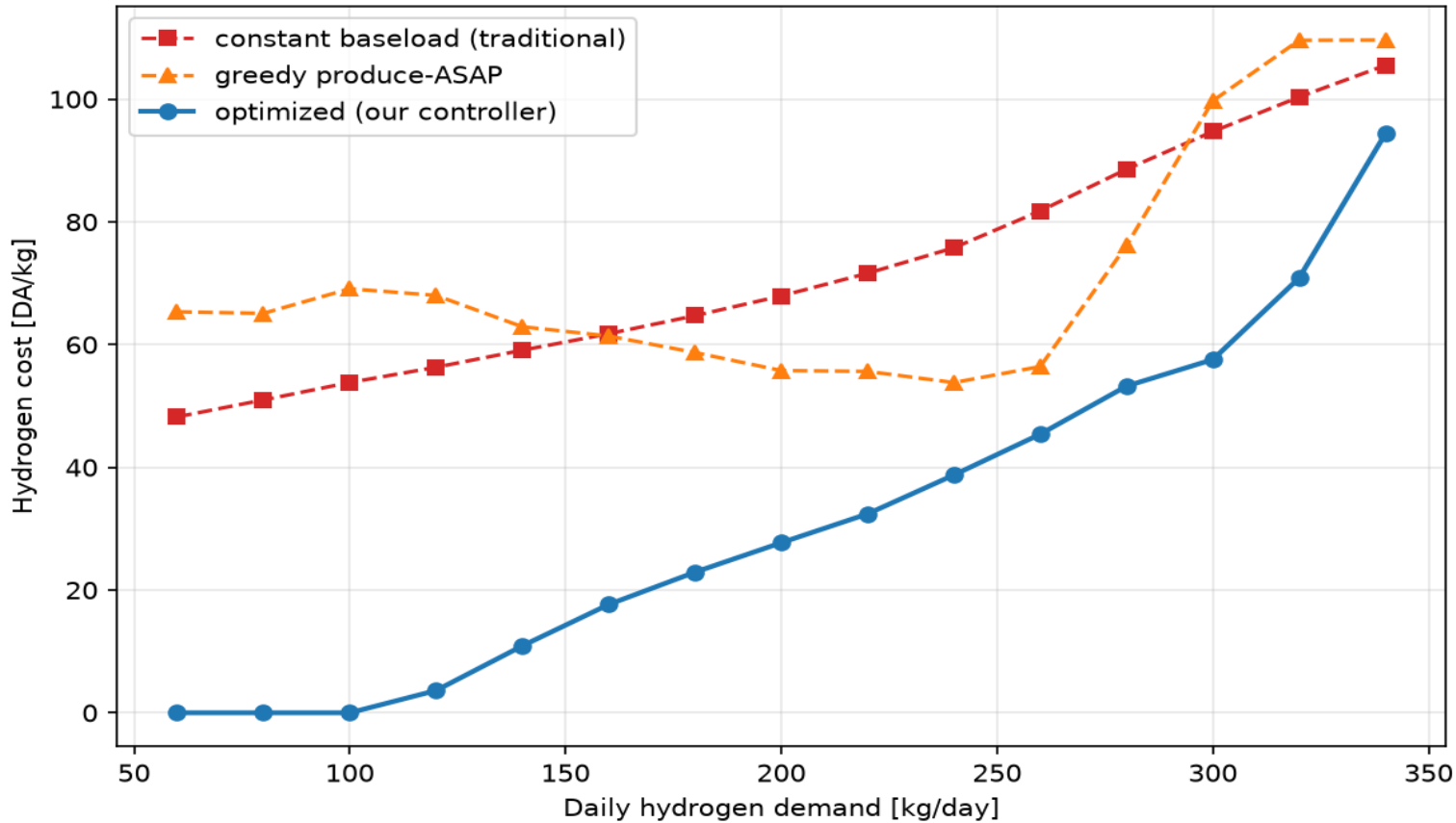
≈ 27.8

DA/kg — clear summer, 200 kg/day

- Full power on cheap night grid (1.21 DA).
- Setpoint tracks the PV mid-day → near-zero grid import.
- Backs off / shuts down through the 8.11 DA peak.
- Cloudy day shows real 15-min cloud transients — the optimiser still finds the cheap windows.

50–59 % cheaper than time-blind operation

Smart dispatch vs traditional operation — Ghardaïa clear summer day
(real CAMS+PySAM PV, same demand, same CREG tariff)



–59%

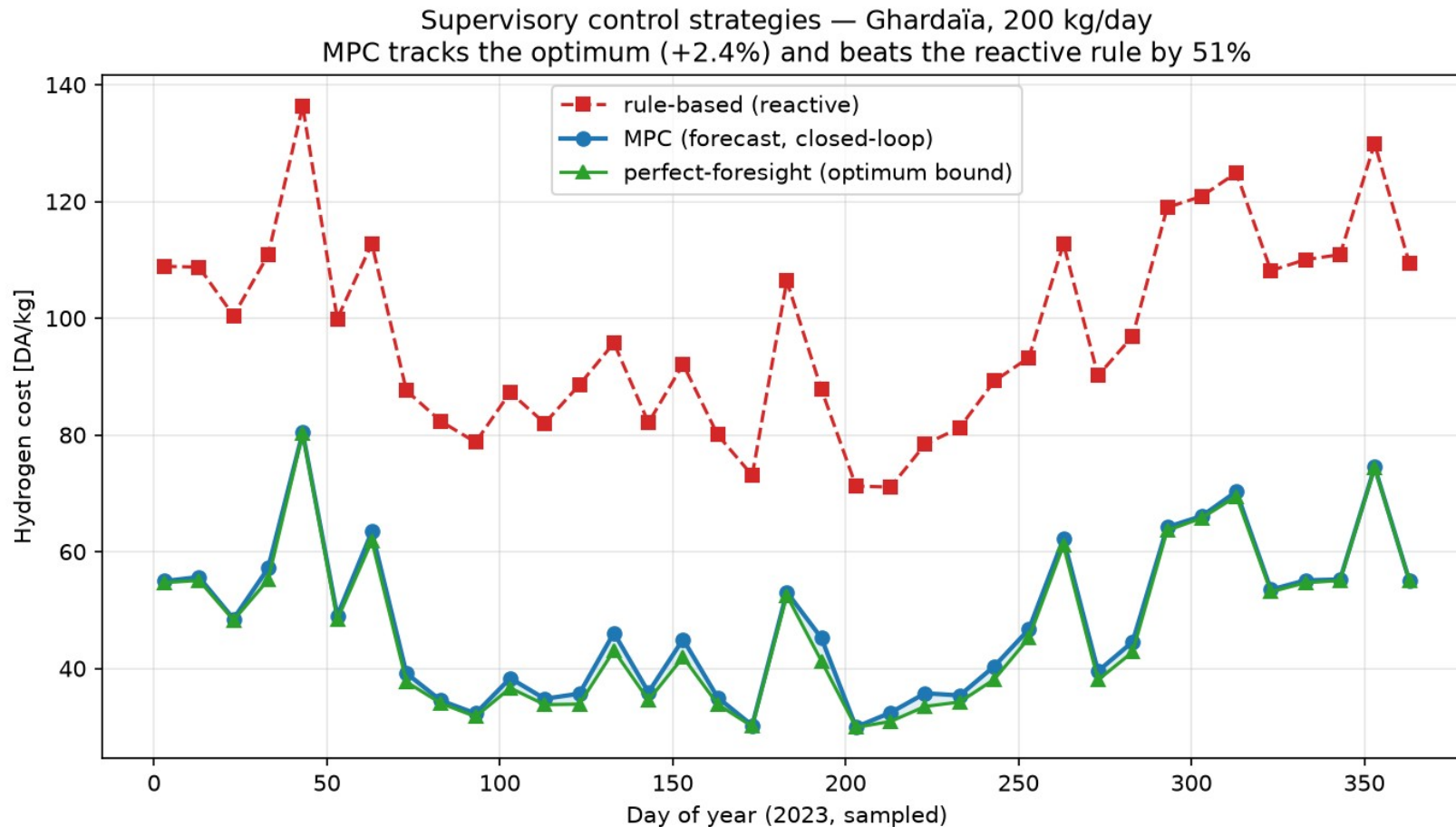
vs constant baseload

–51%

vs greedy ASAP

- Same hydrogen, same plant, same tariff — only the dispatch timing differs.
- Up to 100 % cheaper at low demand fully covered by solar.
- Naive rules degrade as demand rises; the optimiser stays cheapest across the whole range.

MPC nearly matches the optimum, beats reactive control



+2.4%

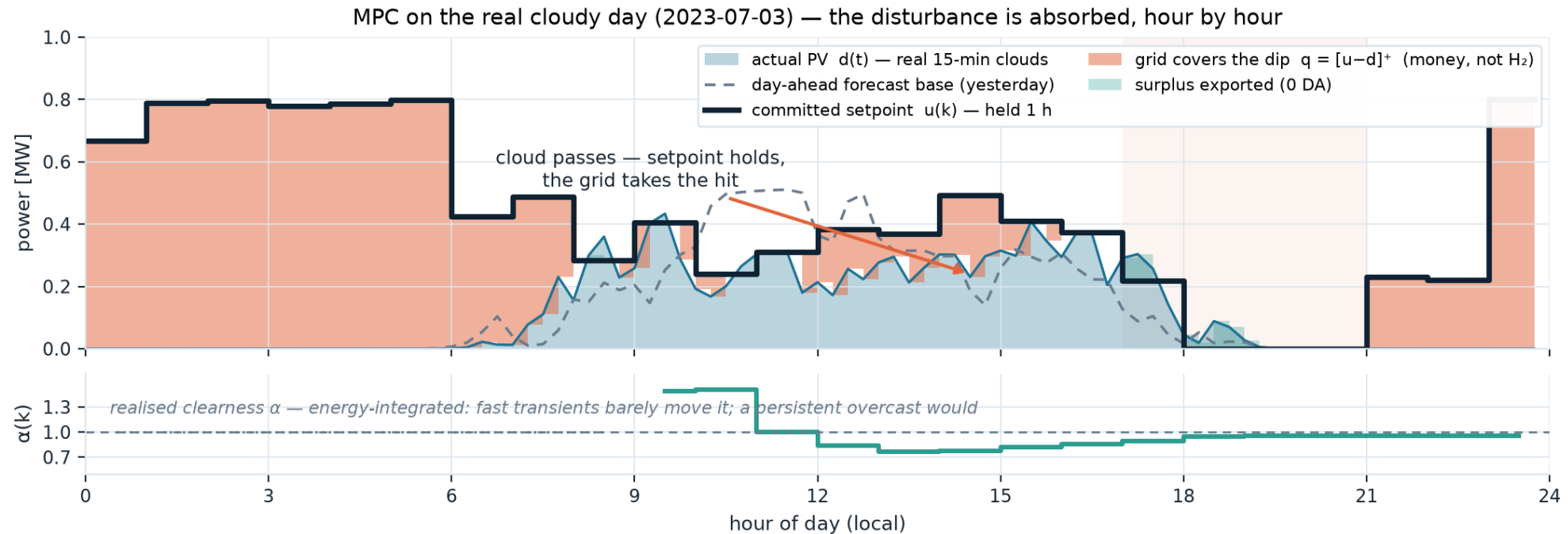
MPC gap to perfect-foresight

-51%

MPC vs reactive rule

Closed-loop MPC recovers $\approx 98\%$ of an unreachable optimum and roughly halves the cost of reactive control — across the year, with imperfect forecasts.

What a passing cloud actually does



WITHIN THE HOUR — PHYSICS

$u(k)$ is held; the grid absorbs every dip instantly: $q = [u - d]^+$. A cloud costs money, never hydrogen — $g(u)$ does not see the weather.

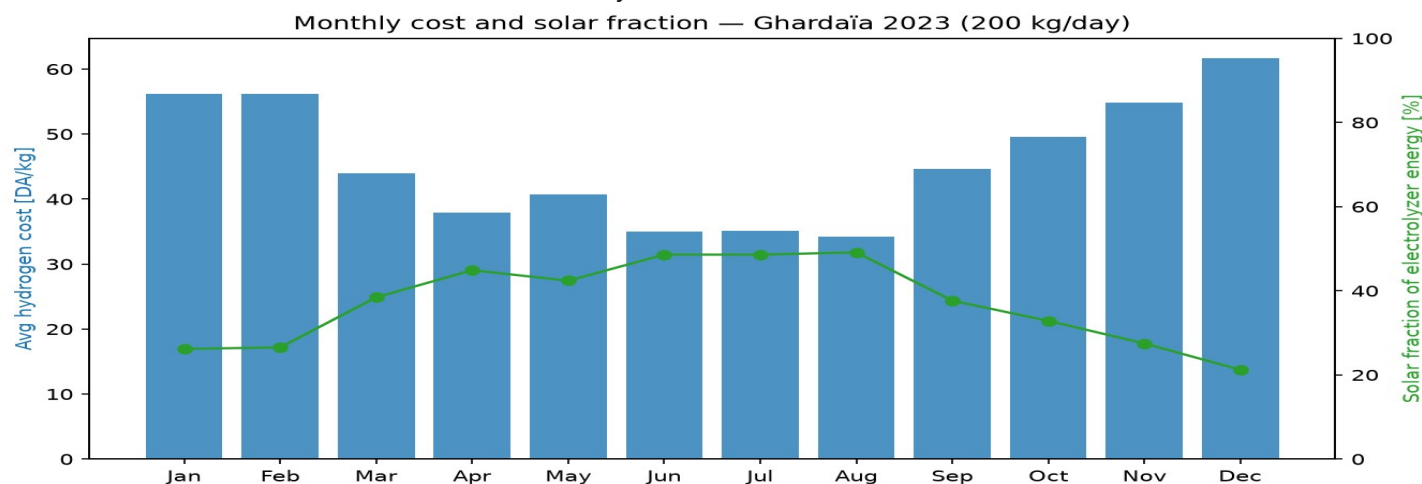
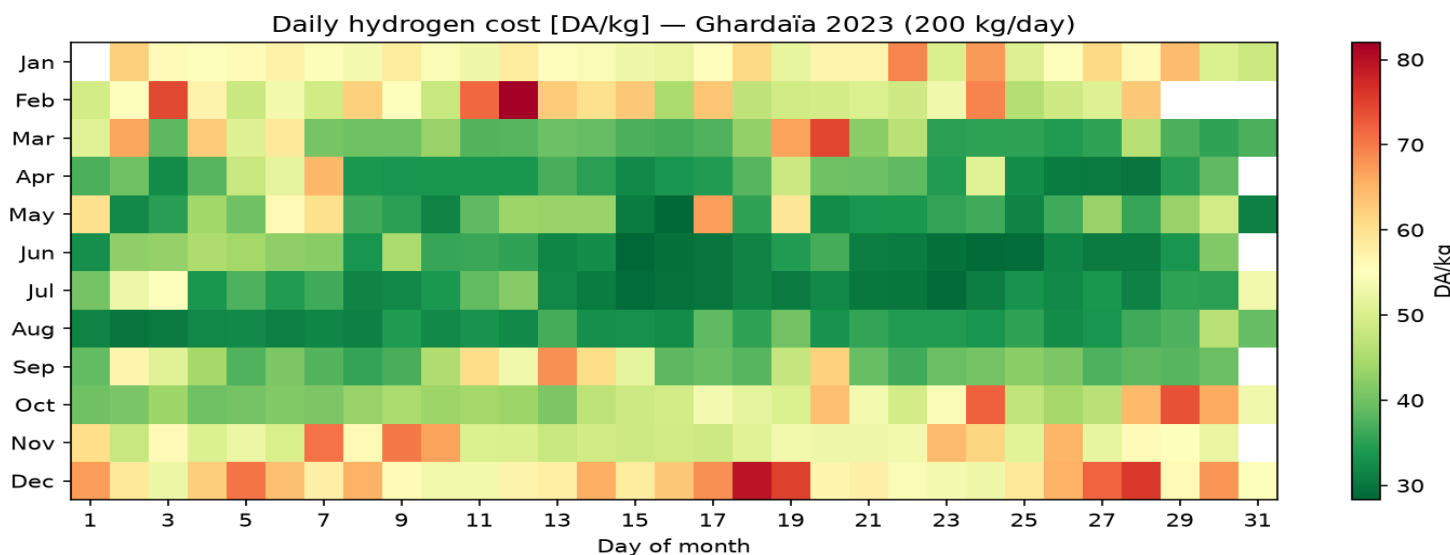
AT THE HOUR — FEEDBACK

Realised energy updates the clearness $\alpha(k)$, and the rest of the day is re-optimised. α integrates energy: fast transients are filtered out; a persistent overcast shifts the whole plan.

THE PRICE — SMALL

This real cloudy day: 200 kg delivered, 55.6 vs 54.4 DA/kg — only +2.1 % above perfect foresight. Year average +2.4 %; worst day $\approx$ +10 %.

A whole year: cheap green summers, dear winters



72.8 t

H₂ / year

364/364

days demand met

45.8

DA/kg avg (28–82)

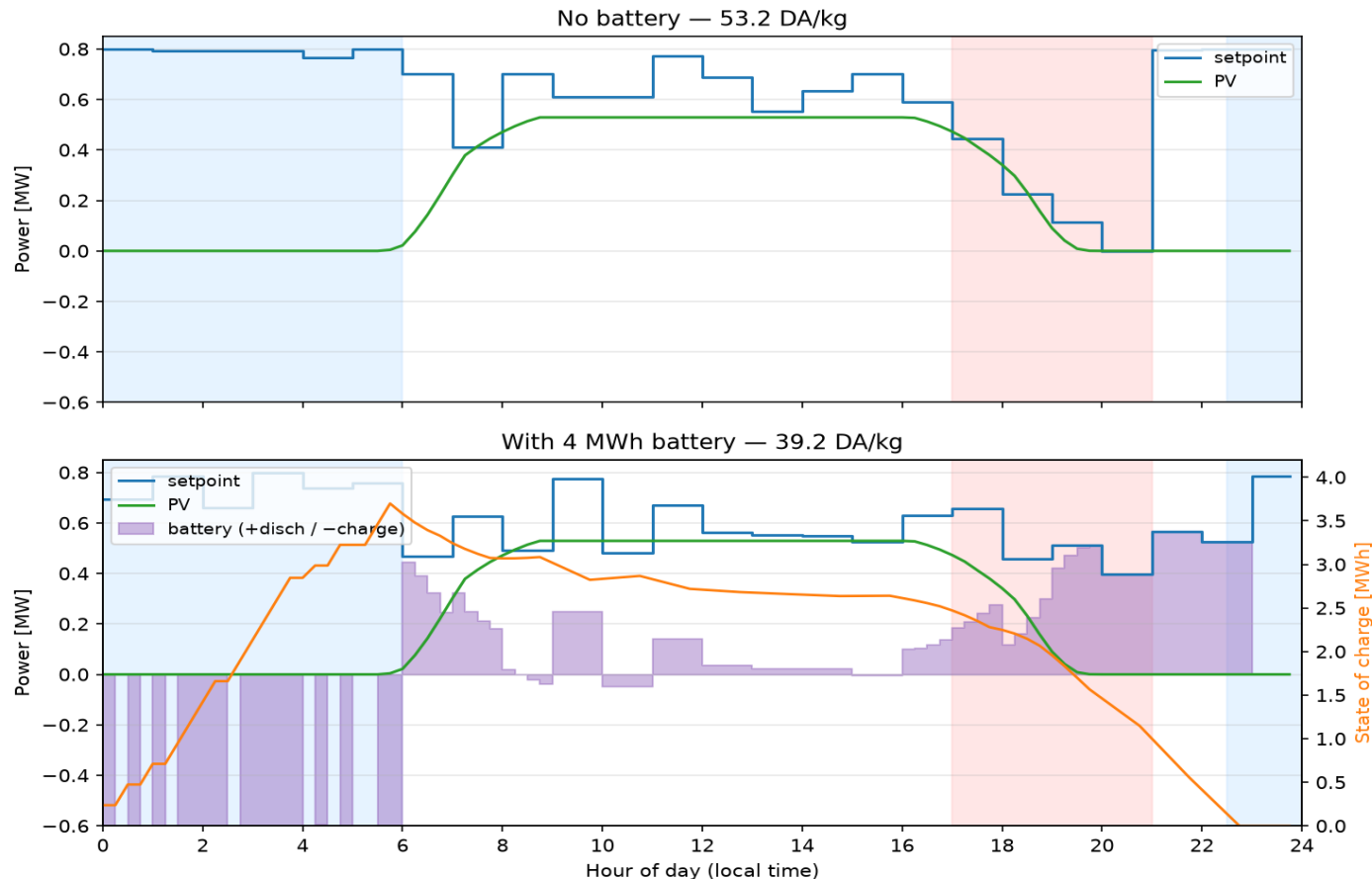
37 %

solar-powered

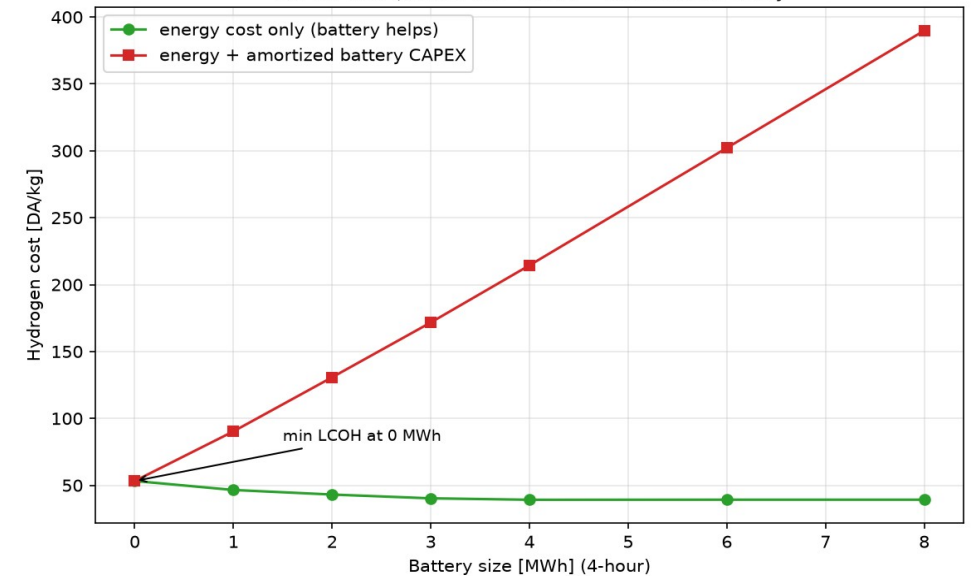
- 364 days simulated end-to-end on real PV.
- Total 3.33 M DA/yr; strong seasonal pattern.
- Summer $\approx$ 28 DA/kg, winter $\approx$ 82 DA/kg — the calendar heatmap shows it at a glance.

A battery helps the bill but not the business case

Battery dispatch — Ghardaïa clear summer, 280 kg/day (real data)
blue night = cheap power, red = 17–21 h peak



Battery sizing — Ghardaïa clear summer, 280 kg/day
CAPEX 250 \$/kWh, CRF 0.133 → 12.3 DA/kWh/day



- With a 4 MWh battery, energy cost falls 53.2 → 39.2 DA/kg (−26 %): it charges on cheap night power + surplus solar and discharges through the peak — real control behaviour.
- But with amortised CAPEX (250 \$/kWh), LCOH is minimised at 0 MWh — storage is not cost-justified under the CREG tariff. Break-even ≈ 20 \$/kWh ($\approx 12\times$ below today).

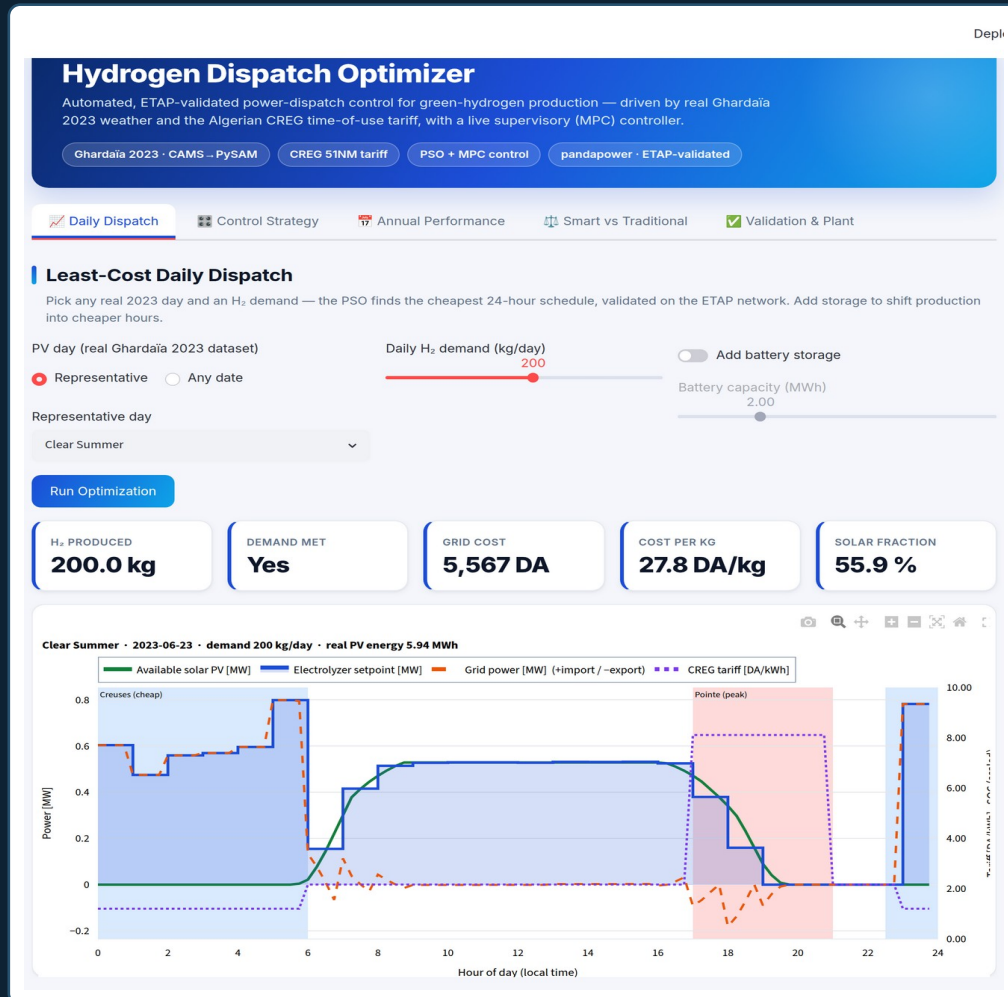
IV

Deliverables, Demo & Conclusion

The live dashboard, and what it all adds up to.

Presented by BOUDJADJA Mohamed Akrem

The controller, live



- A Streamlit dashboard with five tabs: daily dispatch, control strategy, annual performance, smart-vs-traditional, and validation.
- Pick any real 2023 day and an H₂ demand; the PSO runs live and returns the least-cost 24-hour schedule.
- Every run is checked on the ETAP-validated network — feasibility is shown alongside the cost.
- An optional battery toggle overlays the LP-optimal storage trajectory.

Backup if the live demo fails: this screenshot is the Daily-Dispatch tab on the clear-summer day (200 kg/day → ≈ 27.8 DA/kg).

CONCLUSION

What we delivered

- ✓ An ETAP-validated, automatable plant model (24/24 checks).
- ✓ A real-data pipeline: CAMS+ERA5 → PySAM and the real CREG tariff, full 2023.
- ✓ Least-cost dispatch via PSO, and a supervisory MPC controller with feedback.
- ✓ Quantified value: 50–59 % cheaper than time-blind operation; MPC within ≈ 2.4 % of the optimum.
- ✓ A full-year study (72.8 t/yr, 45.8 DA/kg) and an honest storage-feasibility finding.
- ✓ A live deliverable: an interactive dashboard on the validated engine.

In one line: a validated, automated supervisory controller that makes hydrogen for about half the cost of running the plant blindly — and you can watch it work.

Limitations & future work

LIMITATIONS

- Single weather year (2023) and one site — robustness across years/sites not yet swept.
- The network is comfortably within limits, so voltage/loading constraints rarely bind at this PV size.
- Forecast model is a simple persistence + clearness correction, not a learned predictor.
- Decisions are hourly; sub-hourly control only matters once storage is added.

FUTURE WORK

- Sensitivity / robustness across multiple weather years and sites.
- Size the PV larger so network limits activate — then constraints truly drive the control.
- Learned PV forecasting; a policy-based 15-min controller with storage.
- Multi-objective dispatch (cost vs CO₂) and surrogate-model acceleration.

THANK YOU

We welcome your questions.

BOUBIDI Youcef · BOUDJADJA Mohamed Akrem

M2 Automation · University of Boumerdès · July 4, 2026

All code, data and figures are versioned and reproducible (one script per figure).

Backup slides follow ►

APPENDIX

Backup slides for jury questions

Validation · network · tariff · PSO · MPC · electrolyzer · resolution · storage · reproducibility

Minute-by-minute plan (rehearsal sheet — not presented)

0:00	P1	Title & hook — “a controller that roughly halves the cost of green hydrogen — shown running live.”
0:30	P1	Agenda + who presents what (30 s, don't dwell)
1:00	P1	The product — 3× diesel per kg · 97 Mt industry · the colours · electricity = the cost
2:00	P1	Why now (reserves $\approx$ 50 yr, CBAM 2026, solar -90%) → three strategies (141 / 117 kg / 27.8) → the four promises
5:15	P1	Part II — the method in one picture (4 stages, real thumbnails) → real data & the official tariff
7:15	P1	Validation vs the industry reference (24/24) → electrolyzer physics (54 kWh/kg, 10 % turndown)
9:00	P1	① the problem: formulation + formal statement (state-space, Hammerstein, least-cost J) → HANDOVER
10:00	P2	PSO — concept & true update rule → convergence (3 seeds, same $\approx 5,565$ DA optimum)
11:30	P2	③ the loop — MPC named at last: block diagram in equations + receding horizon → the law in one line
13:30	P2	Results — day plan → vs traditional → MPC (+2.4 %, -51%) → cloud anatomy (+2.1 %) → annual → battery
17:20	P2	Demo — dashboard live (~2 min: slider → optimise → control tab → validation; screenshot fallback ready)
19:10	P2	Conclusion → limitations → thank you at 20:00

Running long? Compress in this order: battery → cloud anatomy → hydrogen 101. Never compress: validation, the MPC comparison, the conclusion.

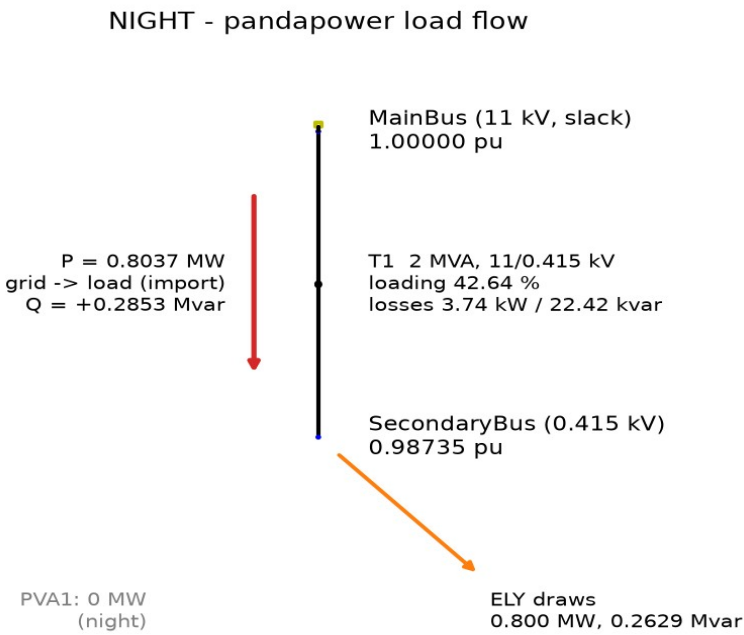
ETAP ↔ pandapower: 24/24 checks

Case	Quantity	ETAP	pandapower	$ \Delta $	Tol.	
SUNNY	SecondaryBus voltage [pu]	0.99042	0.99042	4.3e-06	± 0.01	PASS
NIGHT	SecondaryBus voltage [pu]	0.98735	0.98735	2.9e-06	± 0.01	PASS
PV_EXPORT	SecondaryBus voltage [pu]	1.0026	1.0026	1.1e-06	± 0.01	PASS
SUNNY	T1 loading [%]	18.87	18.874	3.5e-03	± 2 pts	PASS
NIGHT	T1 loading [%]	42.64	42.644	3.8e-03	± 2 pts	PASS
PV_EXPORT	T1 loading [%]	26.71	26.71	9.7e-09	± 2 pts	PASS
SUNNY	T1 real losses [kW]	0.732	0.73166	3.4e-04	± 2 %	PASS
NIGHT	T1 real losses [kW]	3.737	3.7352	1.8e-03	± 2 %	PASS
PV_EXPORT	T1 real losses [kW]	1.459	1.4578	1.2e-03	± 2 %	PASS
SUNNY	T1 reactive losses [kvar]	4.39	4.3921	2.1e-03	± 2 %	PASS
NIGHT	T1 reactive losses [kvar]	22.421	22.422	1.2e-03	± 2 %	PASS
PV_EXPORT	T1 reactive losses [kvar]	8.752	8.751	1.0e-03	± 2 %	PASS
SUNNY	Grid (slack) P [MW]	0.2665	0.26653	3.2e-05	± 2 %	PASS
NIGHT	Grid (slack) P [MW]	0.8037	0.80374	3.5e-05	± 2 %	PASS
PV_EXPORT	Grid (slack) P [MW]	-0.5328	-0.53274	5.8e-05	± 2 %	PASS
SUNNY	Grid (slack) Q [Mvar]	0.2672	0.26729	9.2e-05	± 2 %	PASS
NIGHT	Grid (slack) Q [Mvar]	0.2852	0.28532	1.2e-04	± 2 %	PASS
PV_EXPORT	Grid (slack) Q [Mvar]	0.0088	0.008751	4.9e-05	± 2 %	PASS
SUNNY	ELY drawn P [MW]	0.8	0.8	0.0e+00	± 2 %	PASS
NIGHT	ELY drawn P [MW]	0.8	0.8	0.0e+00	± 2 %	PASS
PV_EXPORT	ELY drawn P [MW]	0	0	0.0e+00	= 0	PASS
SUNNY	ELY drawn Q [Mvar]	0.2628	0.2629	1.0e-04	± 2 %	PASS
NIGHT	ELY drawn Q [Mvar]	0.2628	0.2629	1.0e-04	± 2 %	PASS
PV_EXPORT	ELY drawn Q [Mvar]	0	0	0.0e+00	= 0	PASS

- Three operating points $\times$ eight quantities = 24 checks.
- Voltages within ± 0.01 pu; transformer loading within 2 pts; flows/losses within 2 %.
- PV-export case reproduces the reverse-power-flow sign (grid P = -0.533 MW).
- Newton-Raphson solver, matching ETAP's.
- Every number:
results/tables/validation_pandapower_vs_etap.csv, regenerated by one script.

Single-line model & key parameters

Utility grid U1	11 kV slack, 500 MVA _{sc} , X/R = 10
Transformer T1	2 MVA, 11 / 0.415 kV, Z = 6.25 %, X/R = 6
Electrolyzer ELY	0.800 MW, pf 0.95, constant-power load
PV array PVA1	≈ 600 kWp DC / 630 kVA inv → 534 kW AC full sun
Limits	V 0.95–1.05 pu · transformer ≤ 100 %



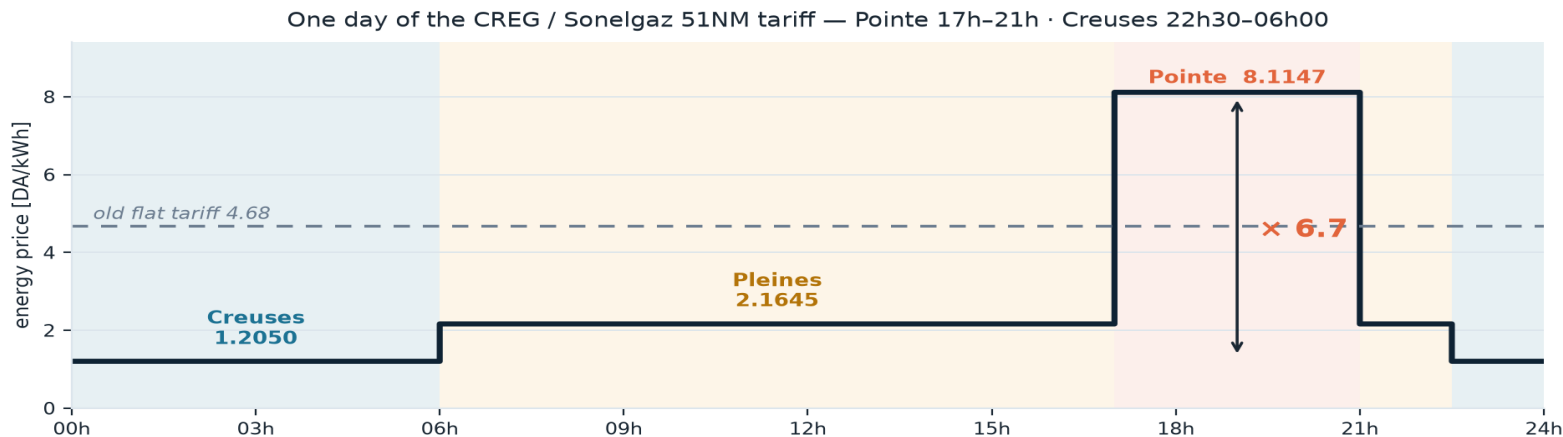
NIGHT case: PV off, grid supplies the full 0.8 MW load.

The CREG 51NM time-of-use tariff

- Official Algerian MT schedule (code 51NM) — exact energy rates: 1.2050 (creuses) / 2.1645 (pleines) / 8.1147 (pointe) DA/kWh.
- Windows: pointe 17h00–21h00 · pleines 06h00–17h00 & 21h00–22h30 · creuses 22h30–06h00.
- Prime de puissance 4.37 DA/kW/month on subscribed power — dispatch-independent, so excluded from the optimisation; imported energy is what the controller can move.
- UTC weather aligned to local time; hourly decisions sample the hour start (22h00–23h00 billed pleines — conservative).

6.7×

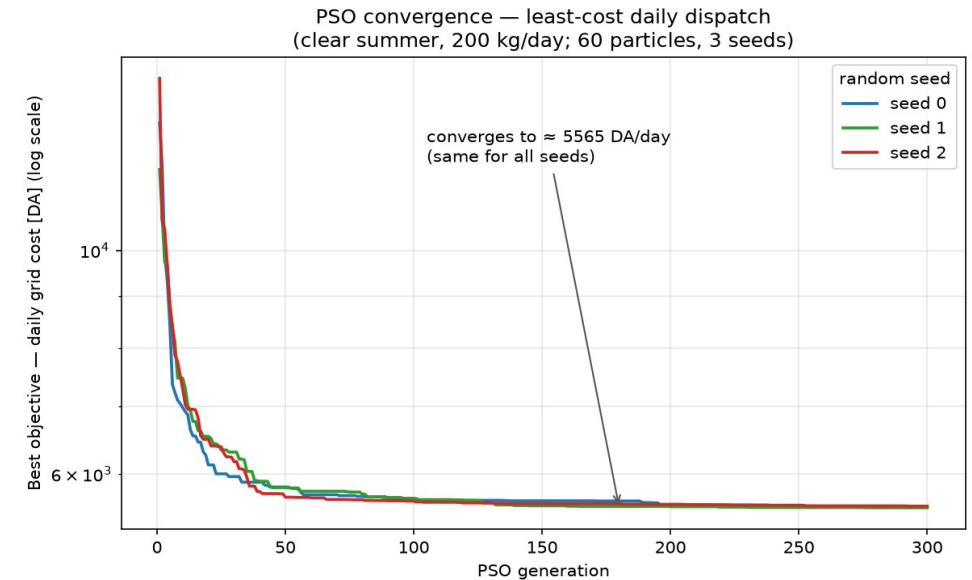
peak-to-night ratio — the spread that makes timing a real decision



One day of the official grille, to the centime — the dashed line is the old flat 4.68 DA/kWh tariff the study replaced.

PSO — formulation & settings

- Decision vector: 24 hourly setpoints $\in \{0\} \cup [0.08, 0.80]$ MW — searched in $[0, 0.8]$, snapped by a repair operator (below turndown $\rightarrow$ off).
- Objective: $J = \sum p \cdot [u - d]^+ \cdot \Delta t$ (15-min accounting) + 10^7 DA/kg $\times$ any demand shortfall.
- Update: $v_i \leftarrow w \cdot v_i + c_1 r_1 (p_i - x_i) + c_2 r_2 (g - x_i)$ with $r_1, r_2 \sim U(0,1)$; $|v| \leq 20\%$ of range; pymoo adapts w, c_1, c_2 online (0.9 / 2.0 / 2.0 start).
- Runs: day plan 60 particles $\times$ 300 generations (converged by ≈ 150); inside the MPC loop 40×120 per re-plan (speed).
- 3 random seeds reach the same $\approx 5,565$ DA optimum $\rightarrow$ reproducible. Gradient-free $\rightarrow$ handles the non-linear g and the discontinuous tariff directly.



MPC — forecast, horizon, feedback

- Day-ahead forecast: yesterday's PV profile as persistence, corrected intraday by how clear today has actually been so far.
- Receding horizon: at hour h , re-optimize hours $h \dots 24$ (PSO) to meet the remaining H_2 demand at least cost.
- Apply only hour h 's setpoint; discard the rest; advance and re-plan when the real PV arrives.
- Benchmarks: perfect-foresight (optimize the known day = lower bound) and rule-based (constant rate = reactive).
- Across the year: MPC $\approx +2.4\%$ vs the bound, -51% vs the rule — disturbance rejection with feedback.

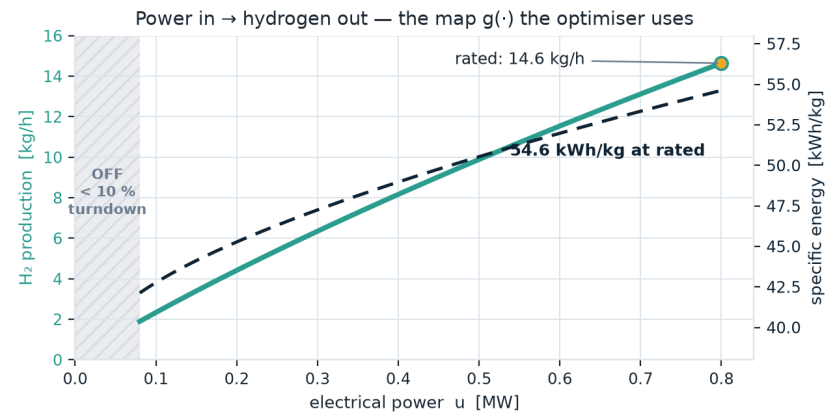
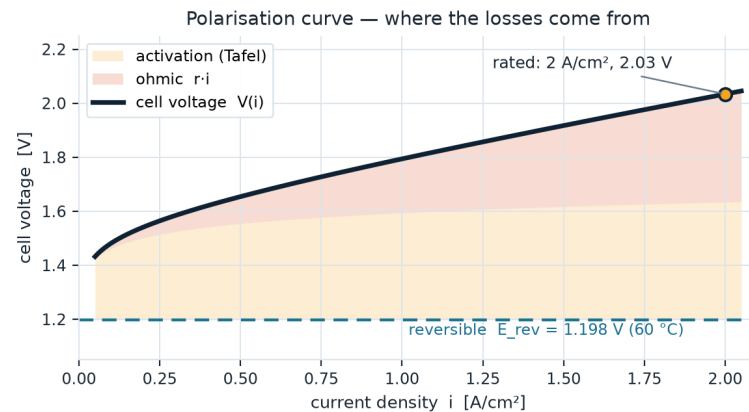
+2.4% / -51%

MPC vs optimum / vs rule

*Code: `src/mpc.py` — `mpc_dispatch()`,
`rule_based_dispatch()`.*

PEM electrolyzer model

- Cell voltage $V(i) = E_{\text{rev}} (1.198 \text{ V at } 60^\circ\text{C}) + \text{Tafel activation } (RT/\alpha F) \cdot \ln(i/i_0) + \text{ohmic } r \cdot i$.
- Hydrogen from Faraday's law: $\dot{m} = \eta_F \cdot n_{\text{cells}} \cdot I / (zF) \cdot M_{\text{H}_2}$ — with $\eta_F = 0.99$, 200 cells.
- Stack sized for exactly 0.800 MW at 2 A/cm^2 ($\approx 2.03 \text{ V/cell}$); operating window 10 %–100 %.
- Part-load is MORE efficient (lower V at lower i) — the dashed specific-energy curve; the optimiser exploits it within the schedule.



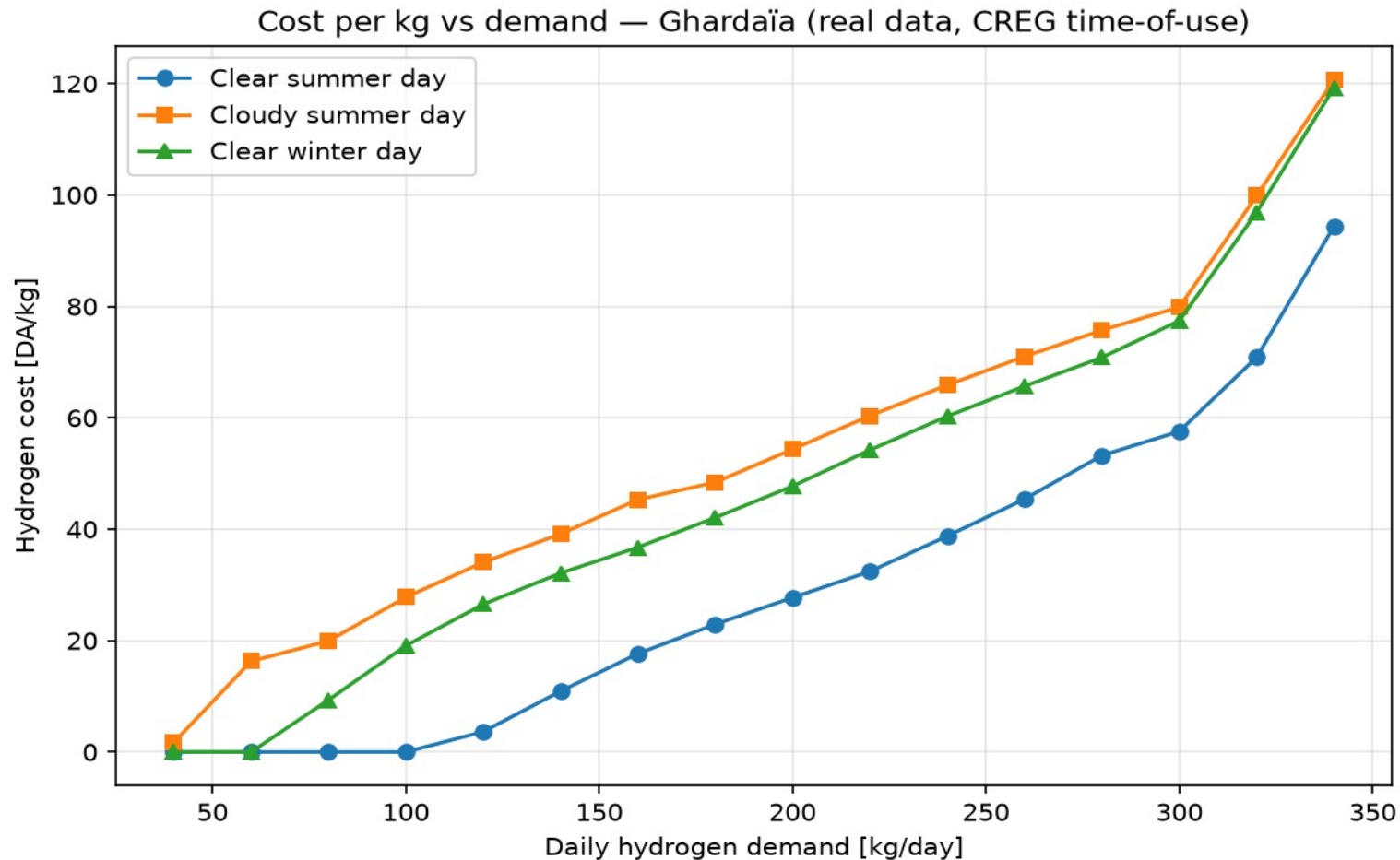
≈ 54

kWh per kg H_2 at rated

14.6

kg H_2 /h at 0.8 MW

Cost per kg vs daily demand



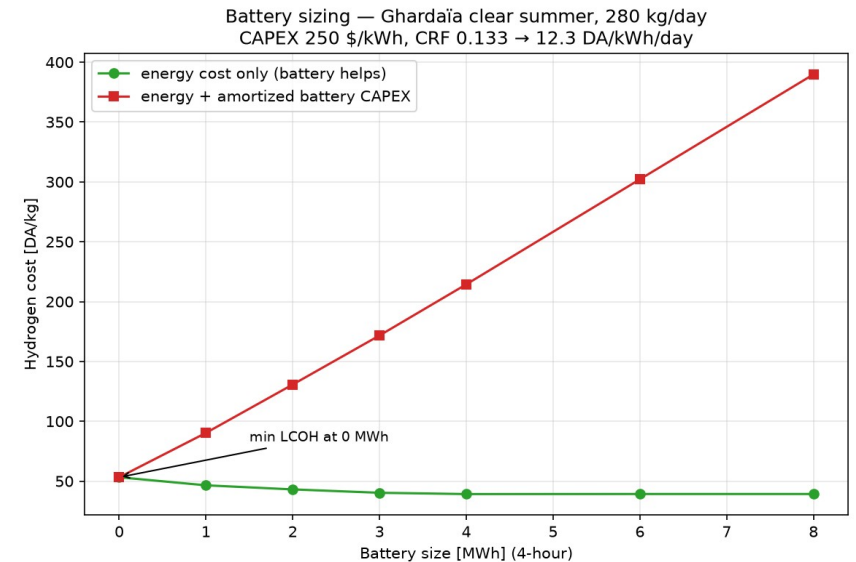
- Low demand can be met almost entirely from free solar → near-zero cost.
- A clear 'knee' where demand outgrows the solar window and grid import (incl. some peak) becomes unavoidable.
- Clear-summer cheapest; cloudy-summer and clear-winter cost more for the same demand.
- This curve is what the dashboard recomputes live when you move the demand slider.

Hourly decisions, 15-minute accounting

- Data, PV, energy, cost and network feasibility are all evaluated at the native 15-minute resolution of the real series.
- The dispatch decision is hourly — 24 setpoints, not 96.
- Why: hourly commitment is the power-industry day-ahead standard and keeps the PSO low-dimensional and reproducible (24 vs 96 variables).
- Sub-hourly control adds negligible value while prices are hourly and there is no storage.
- It becomes worthwhile with the Stage-2 battery, where a policy-based 15-min controller is the planned upgrade.

Decomposed dispatch: PSO + LP

- The storage problem is decomposed, not solved jointly.
- PSO searches only the 24 hourly electrolyzer setpoints (the non-linear, demand-constrained part).
- For each candidate, the battery is operated optimally by a linear program (scipy HiGHS): charge/discharge/SOC that minimise grid cost vs the time-of-use price, subject to power and energy limits.
- Why: a joint 48-variable PSO converged unreliably (non-monotonic sizing). The decomposition gives provably-optimal storage per candidate, reproducible and monotone results, and a low-dimensional PSO.



Data provenance & software stack

Solar irradiance	CAMS Radiation Service v4.6 (Copernicus), satellite-derived, 15-min
Temperature / wind	ERA5 reanalysis (Copernicus CDS), 2023, nearest cell to Ghardaïa
PV model	NREL SAM via PySAM — single-axis tracking, 15-min AC series
Tariff	Algerian CREG medium-voltage time-of-use (code 51NM)
Network / load flow	pandapower (Newton-Raphson), validated against ETAP 20.6
Optimiser	pymoo / PSO; battery LP via scipy HiGHS
Apps	Streamlit dashboard (daily · control · annual · baselines · validation)

Every figure in this deck is regenerated by a single script from the real data — the study is fully reproducible.

Where this work sits (state of the art)

Professional load-flow tools (ETAP)

Accurate and trusted, but GUI-bound — not easily scripted for batch optimisation over a whole year.

GAP: automation

Green-H₂ dispatch & techno-economics

Rich literature on sizing and economics; fewer works close the loop with a real network feasibility check each step.

GAP: network-aware

Metaheuristics for energy scheduling

PSO / GA widely used for unit commitment; we apply PSO to a non-linear electrolyzer under a discontinuous tariff.

USED: PSO engine

MPC for supervisory energy control

Established in microgrids; we apply receding-horizon MPC to electrolyzer dispatch with a measured weather disturbance.

OUR FRAMING

Our gap, in one line: an automated, ETAP-validated dispatch optimiser with a feedback (MPC) controller and a live demonstration — on real data.